Solution Requirements

Date	05 July 2026
Team ID	
Project Name	Green-Predict-Pro
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Functional Requirements (FR) define specific behaviors, functions, or features of the system (what the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (how the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Farmers, researchers and admins log in using email and password with hashed credentials.	Medium
2	Authorization levels	Three roles – Farmer (input & view result), Researcher (analytics), Admin (manage users & dataset).	High
3	External interfaces	Weather API (optional) to auto-fill temperature/humidity/rainfall for a given location.	Medium
4	Transactions processing	Accept crop-parameter inputs, run trained ML model and return the recommended crop with confidence score.	High
5	Reporting	Downloadable PDF report of inputs, recommendation and suitability score for each session.	Medium
6	Business rules	Values must fall within valid ranges (e.g., pH 0-14, humidity 0-100%). Invalid inputs are rejected with a clear message.	High
7	Compliance to laws / regulations	No personal data beyond email is stored; app follows Indian IT Act & general data-privacy best practices.	Medium
8	Other	Multilingual UI (English + regional language) and offline-friendly form.	Low

SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Recommendation returned within 2 seconds of form submission.	Response time < 2 seconds.
2	Scalability	System should serve concurrent farmer users during peak sowing seasons.	Supports up to 5,000 concurrent users.
3	Security & Data Privacy	Encrypted passwords, HTTPS-only, no sensitive PII collected.	bcrypt hashing, TLS 1.2+.
4	Reliability & Availability	Service available around the clock during cropping seasons.	99.5% uptime per month.
5	Usability & Accessibility	Simple form suitable for low-literacy users; mobile responsive.	Complies with WCAG 2.1 AA.
6	Other	Maintainability – modular Flask code with clearly separated ML pipeline.	Codebase documented; retraining possible in under 1 day.